Fluids and Blood Transfusion MCQ – Chapter 7

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 3: Quintessential in Anesthesia (Chapter 7)

50 Questions • 30 Minutes

Instructions:
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

Questions

Q1. According to the textbook, fluids are broadly divided into which two categories? **A.** Isotonic and hypertonic **B.** Crystalloids and colloids **C.** Dextrose and saline **D.** Natural and synthetic	**Ans: B**
Q2. What is the composition of Plasmalyte solution as per the textbook? **A.** Na+ 154, Cl- 154 mEq/L **B.** Na+ 140, Cl- 98, K+ 3, Mg2+ 3, Acetate 27, Gluconate 23 mEq/L **C.** Na+ 131, Cl- 111, K+ 5 mEq/L **D.** Na+ 130, Cl- 109, Lactate 28 mEq/L	**Ans: B**
Q3. Normal saline is preferred over Ringer lactate in all of the following conditions EXCEPT: **A.** Hypochloremic metabolic alkalosis **B.** Brain injury **C.** Hyponatremia **D.** Hypokalemia	**Ans: D**
Q4. According to Table 7.1, crystalloids should be replaced in what ratio to blood loss? **A.** 1:1 ratio **B.** 1.5:1 ratio (150 mL crystalloid for 100 mL blood loss) **C.** 3:1 ratio **D.** 2:1 ratio	**Ans: B**
Q5. Why should blood NOT be given through the same drip set as Ringer lactate? **A.** RL is too viscous **B.** RL contains calcium which can cause clotting of blood **C.** RL is hypertonic and can lyse RBCs **D.** RL contains potassium which causes hemolysis	**Ans: B**
Q6. Which of the following is an isotonic crystalloid solution? **A.** 5% Dextrose **B.** Normal saline (0.9% NaCl) **C.** 3% Hypertonic saline **D.** 0.45% NaCl	**Ans: B**
Q7. The sodium content of normal saline (0.9% NaCl) is: **A.** 100 mEq/L **B.** 130 mEq/L **C.** 154 mEq/L **D.** 170 mEq/L	**Ans: C**

Q8. Ringer's lactate solution contains all of the following EXCEPT: **A.** Na+ 131 mEq/L **B.** K+ 5 mEq/L **C.** Lactate 29 mEq/L **D.** Bicarbonate 29 mEq/L	**Ans: D**
Q9. The lactate in Ringer's lactate is converted to bicarbonate in the: **A.** Kidney **B.** Liver **C.** Lung **D.** Muscle	**Ans: B**
Q10. Excessive administration of normal saline can cause: **A.** Metabolic alkalosis **B.** Hyperchloremic metabolic acidosis **C.** Respiratory alkalosis **D.** Hypokalemia	**Ans: B**
Q11. 5% Dextrose is considered isotonic in the bag but acts as a hypotonic solution in vivo because: **A.** It contains sodium **B.** Dextrose is rapidly metabolized, leaving free water that distributes across all compartments **C.** It has high osmolarity **D.** It contains lactate	**Ans: B**
Q12. According to Table 7.1, the intravascular half-life of crystalloids is approximately: **A.** 5 minutes **B.** 30 minutes **C.** 2–4 hours **D.** 24 hours	**Ans: B**
Q13. Colloids differ from crystalloids in that they: **A.** Are freely filtered across capillary membranes **B.** Contain large molecular weight substances that maintain oncotic pressure and stay intravascular longer **C.** Are always synthetic **D.** Cannot be used for volume resuscitation	**Ans: B**
Q14. Which of the following is a natural colloid? **A.** Hydroxyethyl starch (HES) **B.** Dextran **C.** Human albumin **D.** Gelatin (Haemaccel)	**Ans: C**

Q15. The approximate intravascular half-life of infused colloids compared to crystalloids is: **A.** Shorter **B.** Same **C.** Longer (several hours) **D.** Only minutes	**Ans: C**

Q16. Dextran 40 is used primarily for: **A.** Volume expansion only **B.** Improving microcirculation and reducing blood viscosity (rheological effects) **C.** Providing nutrition **D.** Correcting acidosis	**Ans: B**

Q17. A major side effect of dextran infusion is: **A.** Hyperkalemia **B.** Anaphylactic reactions and interference with blood cross-matching **C.** Metabolic alkalosis **D.** Hypertension	**Ans: B**

Q18. Hydroxyethyl starch (HES) solutions can cause: **A.** Improved renal function **B.** Coagulopathy and renal impairment at high doses **C.** Hypernatremia **D.** Metabolic alkalosis	**Ans: B**

Q19. Gelatin-based colloids (e.g., Haemaccel) have which disadvantage? **A.** Very long intravascular half-life **B.** Risk of anaphylactoid reactions and relatively short duration of action **C.** Severe hyperkalemia **D.** Incompatible with all blood products	**Ans: B**

Q20. The daily maintenance fluid requirement for a 70 kg adult using the 4-2-1 rule is approximately: **A.** 50 mL/hr **B.** 110 mL/hr **C.** 150 mL/hr **D.** 200 mL/hr	**Ans: B**

Q21. The 4-2-1 rule for hourly maintenance fluid calculation gives 4 mL/kg/hr for: **A.** First 20 kg **B.** First 10 kg **C.** Body weight above 20 kg **D.** All body weight	**Ans: B**

Q22. Intraoperative fluid therapy must account for: **A.** Maintenance requirements only **B.** Maintenance, deficit (fasting), and ongoing losses (surgical/third space) **C.** Blood loss only **D.** Urine output only	**Ans: B**

Q23. Third-space fluid loss refers to: **A.** Blood loss into the suction canister **B.** Functional loss of fluid into a non-equilibrating space (e.g., bowel wall, traumatized tissues) **C.** Urine output **D.** Sweat loss	**Ans: B**

Q24. For major abdominal surgery, the estimated third-space fluid loss is approximately: **A.** 1–2 mL/kg/hr **B.** 4–8 mL/kg/hr **C.** 6–10 mL/kg/hr **D.** 0.5 mL/kg/hr	**Ans: C**

Q25. The preoperative fasting deficit is calculated as: **A.** Body weight × hours of fasting **B.** Hourly maintenance rate × number of hours NPO **C.** Blood volume × hematocrit **D.** Urine output × hours	**Ans: B**

Q26. Whole blood transfusion contains all components of blood and has a hematocrit of approximately: **A.** 20–25% **B.** 35–40% **C.** 55–60% **D.** 70–80%	**Ans: B**

Q27. According to Table 7.1, colloids in high doses can interfere with clotting primarily by: **A.** Activating platelets **B.** Decreasing Factor VIII level and causing renal dysfunction **C.** Increasing fibrinogen levels **D.** Stimulating thrombin production	**Ans: B**

Q28. One unit of packed RBCs raises hemoglobin by approximately: **A.** 0.5 g/dL **B.** 1 g/dL **C.** 3 g/dL **D.** 5 g/dL	**Ans: B**

Q29. The storage temperature for packed red blood cells is: **A.** Room temperature (20–24°C) **B.** 1–6°C **C.** -20°C **D.** -80°C	**Ans: B**
Q30. Fresh frozen plasma (FFP) is indicated for: **A.** Anemia **B.** Replacement of coagulation factors in patients with coagulopathy **C.** Volume expansion as first choice **D.** Thrombocytopenia	**Ans: B**
Q31. Platelet concentrate transfusion is indicated when platelet count falls below: **A.** 150,000/μL **B.** 100,000/μL **C.** 50,000/μL (or <75,000 with active bleeding/surgery) **D.** 200,000/μL	**Ans: C**
Q32. Platelets are stored at: **A.** 1–6°C **B.** 20–24°C with continuous agitation **C.** -30°C **D.** 37°C	**Ans: B**
Q33. The most common cause of fatal transfusion reaction is: **A.** Allergic reaction **B.** ABO-incompatible (hemolytic) transfusion due to clerical error **C.** Febrile non-hemolytic reaction **D.** Iron overload	**Ans: B**
Q34. Signs of acute hemolytic transfusion reaction under general anesthesia include: **A.** Hypothermia and bradycardia **B.** Unexplained hypotension, fever, hemoglobinuria, and DIC **C.** Hypertension and tachycardia only **D.** Bronchospasm only	**Ans: B**
Q35. The first step in managing suspected hemolytic transfusion reaction is: **A.** Continue transfusion at slower rate **B.** Stop the transfusion immediately and maintain IV access **C.** Administer more fluids through the same line **D.** Give antihistamines and continue	**Ans: B**
Q36. Transfusion-related acute lung injury (TRALI) presents with: **A.** Delayed hemolysis after 7 days **B.** Acute respiratory distress, non-cardiogenic pulmonary edema within 6 hours of transfusion **C.** Chronic iron overload **D.** Urticaria only	**Ans: B**

Q37. Massive blood transfusion is defined as replacement of: **A.** One unit of blood **B.** More than one blood volume (or >10 units PRBCs) in 24 hours **C.** Any amount of FFP **D.** 500 mL of blood	**Ans: B**
Q38. Complications of massive blood transfusion include all EXCEPT: **A.** Hypothermia **B.** Hyperkalemia **C.** Citrate toxicity (hypocalcemia) **D.** Polycythemia	**Ans: D**
Q39. Citrate toxicity from massive transfusion manifests as: **A.** Hyperkalemia **B.** Hypocalcemia (ionized calcium reduction) causing cardiac depression **C.** Hypernatremia **D.** Metabolic alkalosis	**Ans: B**
Q40. Hyperkalemia in stored blood occurs because: **A.** Potassium is added as preservative **B.** Red cell membrane Na-K ATPase pump fails during storage, leaking potassium into plasma **C.** Citrate releases potassium **D.** Platelets release potassium	**Ans: B**
Q41. Febrile non-hemolytic transfusion reaction is caused by: **A.** ABO incompatibility **B.** Antibodies against donor leukocyte antigens or cytokines in stored blood **C.** Bacterial contamination **D.** Volume overload	**Ans: B**
Q42. Transfusion-associated circulatory overload (TACO) is characterized by: **A.** Hypotension and bradycardia **B.** Pulmonary edema, hypertension, dyspnea due to volume overload **C.** Hemolysis **D.** Fever only	**Ans: B**
Q43. Infectious complications of blood transfusion include all EXCEPT: **A.** Hepatitis B and C **B.** HIV **C.** Malaria **D.** Tuberculosis (commonly)	**Ans: D**
Q44. Autologous blood transfusion refers to: **A.** Transfusing blood from a family member **B.** Transfusing the patient's own previously collected or salvaged blood **C.** Using synthetic blood substitutes **D.** Directed donor transfusion	**Ans: B**

Q45. Types of autologous blood transfusion include all EXCEPT: **A.** Preoperative autologous donation (PAD) **B.** Acute normovolemic hemodilution (ANH) **C.** Intraoperative cell salvage **D.** Emergency crossmatch from blood bank	**Ans: D**
Q46. In acute normovolemic hemodilution (ANH), blood is removed preoperatively and replaced with: **A.** Nothing **B.** Crystalloids or colloids to maintain normovolemia **C.** Whole blood from donors **D.** Platelets only	**Ans: B**
Q47. Intraoperative cell salvage is contraindicated in: **A.** Cardiac surgery **B.** Surgery for malignancy (relative) and infected surgical fields **C.** Orthopedic surgery **D.** Vascular surgery	**Ans: B**
Q48. The trigger for blood transfusion in a healthy patient is generally a hemoglobin of: **A.** 10 g/dL **B.** 7–8 g/dL **C.** 12 g/dL **D.** 5 g/dL	**Ans: B**
Q49. The anticoagulant used in blood storage bags is: **A.** Heparin **B.** CPDA-1 (Citrate-Phosphate-Dextrose-Adenine) **C.** Warfarin **D.** EDTA	**Ans: B**
Q50. Blood must be transfused through a filter of pore size: **A.** 20 micrometers **B.** 170–260 micrometers (standard blood filter) **C.** 0.22 micrometers **D.** 500 micrometers	**Ans: B**

Answer Key & Explanations

Q1. Answer: B — Crystalloids and colloids

Fluids are divided into crystalloids and colloids as stated at the beginning of Chapter 7.

Topic: Fluids | Ref: Ch 7, Page 71

Q2. Answer: B — Na+ 140, Cl- 98, K+ 3, Mg2+ 3, Acetate 27, Gluconate 23 mEq/L

Plasmalyte composition: Na+ 140, Cl- 98, K+ 3, Mg2+ 3, Acetate 27, Gluconate 23 mEq/L. Acetate and gluconate are metabolized to generate bicarbonate.

Topic: Crystalloids | Ref: Ch 7, Page 72

Q3. Answer: D — Hypokalemia

As per the textbook, NS is preferred over RL for treating: hypochloremic metabolic alkalosis, brain injury (Ca2+ in RL can increase neuronal injury), and hyponatremia.

Topic: Crystalloids | Ref: Ch 7, Page 72

Q4. Answer: B — 1.5:1 ratio (150 mL crystalloid for 100 mL blood loss)

As per Table 7.1, crystalloids should be replaced in a ratio of 1.5:1. For example, blood loss of 100 mL should be replaced with 150 mL of crystalloid.

Topic: Fluids | Ref: Ch 7, Page 71

Q5. Answer: B — RL contains calcium which can cause clotting of blood

As RL contains calcium, blood should not be given through the same drip set as calcium can initiate coagulation of stored blood.

Topic: Crystalloids | Ref: Ch 7, Page 72

Q6. Answer: B — Normal saline (0.9% NaCl)

Normal saline (0.9% NaCl) is an isotonic crystalloid with an osmolarity of 308 mOsm/L, close to plasma osmolarity.

Topic: Crystalloids | Ref: Ch 7, Page 71

Q7. Answer: C — 154 mEq/L

Normal saline contains 154 mEq/L of sodium and 154 mEq/L of chloride.

Topic: Crystalloids | Ref: Ch 7, Page 71

Q8. Answer: D — Bicarbonate 29 mEq/L

Ringer's lactate contains Na+ 131, Cl- 111, K+ 5, Ca2+ 2, Lactate 29 mEq/L, pH 6.5. It does not contain bicarbonate directly.

Topic: Crystalloids | Ref: Ch 7, Page 71

Q9. Answer: B — Liver

Lactate is metabolized by the liver to bicarbonate, providing a buffering effect.

Topic: Crystalloids | Ref: Ch 7, Page 71

Q10. Answer: B — Hyperchloremic metabolic acidosis

Large volumes of NS (high chloride 154 mEq/L) can cause dilutional hyperchloremic metabolic acidosis.

Topic: Crystalloids | Ref: Ch 7, Page 71

Q11. Answer: B — Dextrose is rapidly metabolized, leaving free water that distributes across all compartments

After infusion, glucose is rapidly metabolized, leaving free water that distributes evenly across ICF and ECF, effectively acting as a hypotonic solution.

Topic: Crystalloids | Ref: Ch 7, Page 71

Q12. Answer: B — 30 minutes

As per Table 7.1, crystalloids have an intravascular half-life of 30 minutes, so they expand plasma volume for less time compared to colloids.

Topic: Crystalloids | Ref: Ch 7, Page 71 (Table 7.1)

Q13. Answer: B — Contain large molecular weight substances that maintain oncotic pressure and stay intravascular longer

Colloids contain large molecules (proteins/starches) that generate oncotic pressure, keeping fluid in the intravascular space longer than crystalloids.

Topic: Colloids | Ref: Ch 7, Page 72

Q14. Answer: C — Human albumin

Human albumin is the only natural colloid. HES, dextrans, and gelatins are synthetic colloids.

Topic: Colloids | Ref: Ch 7, Page 72

Q15. Answer: C — Longer (several hours)

Colloids have a longer intravascular half-life (hours) compared to crystalloids because large molecules are retained within the vascular space.

Topic: Colloids | Ref: Ch 7, Page 72

Q16. Answer: B — Improving microcirculation and reducing blood viscosity (rheological effects)

Dextran 40 (low molecular weight) improves microcirculation by reducing blood viscosity and preventing rouleaux formation.

Topic: Colloids | Ref: Ch 7, Page 72

Q17. Answer: B — Anaphylactic reactions and interference with blood cross-matching

Dextrans can cause anaphylaxis, interfere with blood cross-matching (rouleaux formation), and impair coagulation at high doses.

Topic: Colloids | Ref: Ch 7, Page 72

Q18. Answer: B — Coagulopathy and renal impairment at high doses

HES can cause dose-dependent coagulopathy (reduced Factor VIII, von Willebrand factor) and nephrotoxicity, especially in critically ill patients.

Topic: Colloids | Ref: Ch 7, Page 72

Q19. Answer: B — Risk of anaphylactoid reactions and relatively short duration of action

Gelatins have a shorter duration than other colloids and carry a risk of anaphylactoid/allergic reactions.

Topic: Colloids | Ref: Ch 7, Page 72

Q20. Answer: B — 110 mL/hr

4-2-1 rule: 4 mL/kg/hr for first 10 kg (40) + 2 mL/kg/hr for next 10 kg (20) + 1 mL/kg/hr for remaining 50 kg (50) = 110 mL/hr.

Topic: Fluid Management | Ref: Ch 7, Page 73

Q21. Answer: B — First 10 kg

The 4-2-1 rule: 4 mL/kg/hr for the first 10 kg, 2 mL/kg/hr for the next 10 kg, and 1 mL/kg/hr for each kg above 20 kg.

Topic: Fluid Management | Ref: Ch 7, Page 73

Q22. Answer: B — Maintenance, deficit (fasting), and ongoing losses (surgical/third space)

Intraoperative fluids must replace maintenance needs, preoperative fasting deficit, and ongoing surgical/third-space losses.

Topic: Fluid Management | Ref: Ch 7, Page 73

Q23. Answer: B — Functional loss of fluid into a non-equilibrating space (e.g., bowel wall, traumatized tissues)

Third-space loss is the redistribution of fluid into a non-functional compartment (traumatized tissue, bowel wall, peritoneal cavity) that does not participate in normal fluid exchange.

Topic: Fluid Management | Ref: Ch 7, Page 73

Q24. Answer: C — 6–10 mL/kg/hr

Third-space losses vary: 1–2 mL/kg/hr for minimal trauma, 4–6 for moderate, and 6–10 mL/kg/hr for major abdominal surgery.

Topic: Fluid Management | Ref: Ch 7, Page 73

Q25. Answer: B — Hourly maintenance rate × number of hours NPO

Fasting deficit = maintenance fluid rate (by 4-2-1 rule) × hours of fasting (NPO period).

Topic: Fluid Management | Ref: Ch 7, Page 73

Q26. Answer: B — 35–40%

Whole blood has a hematocrit of approximately 35–40% and contains RBCs, plasma, platelets, and clotting factors.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

Q27. Answer: B — Decreasing Factor VIII level and causing renal dysfunction

As per Table 7.1, colloids in high doses can interfere with clotting by primarily decreasing Factor VIII level and can also cause renal dysfunction.

Topic: Colloids | Ref: Ch 7, Page 71 (Table 7.1)

Q28. Answer: B — 1 g/dL

One unit of PRBCs raises hemoglobin by approximately 1 g/dL (or hematocrit by 3%) in an average adult.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

Q29. Answer: B — 1–6°C

PRBCs are stored at 1–6°C with a shelf life of 35–42 days depending on the anticoagulant/preservative used.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

Q30. Answer: B — Replacement of coagulation factors in patients with coagulopathy

FFP contains all coagulation factors and is indicated for coagulopathy, DIC, massive transfusion, and warfarin reversal.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

Q31. Answer: C — 50,000/µL (or <75,000 with active bleeding/surgery)

Platelet transfusion is generally indicated when count is <50,000/µL or <75,000–100,000/µL with active bleeding or before surgery.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

Q32. Answer: B — 20–24°C with continuous agitation

Platelets are stored at room temperature (20–24°C) with continuous gentle agitation, with a shelf life of 5 days.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

Q33. Answer: B — ABO-incompatible (hemolytic) transfusion due to clerical error

ABO-incompatible acute hemolytic transfusion reaction, usually due to clerical/identification error, is the most common cause of fatal transfusion reactions.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

Q34. Answer: B — Unexplained hypotension, fever, hemoglobinuria, and DIC

Under GA, signs include unexplained hypotension, tachycardia, fever, hemoglobinuria (dark urine), diffuse bleeding (DIC), and bronchospasm.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

Q35. Answer: B — Stop the transfusion immediately and maintain IV access

The immediate action is to STOP the transfusion, maintain IV access with new tubing, and send the blood bag and patient sample for verification.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

Q36. Answer: B — Acute respiratory distress, non-cardiogenic pulmonary edema within 6 hours of transfusion

TRALI presents with acute hypoxemia, bilateral pulmonary infiltrates, and respiratory distress within 6 hours of transfusion, without circulatory overload.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

Q37. Answer: B — More than one blood volume (or >10 units PRBCs) in 24 hours

Massive transfusion is defined as replacement of more than one blood volume (approximately 70 mL/kg or >10 units PRBCs) within 24 hours.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

Q38. Answer: D — Polycythemia

Massive transfusion complications include hypothermia, hyperkalemia, citrate toxicity (hypocalcemia), coagulopathy, and acid-base disturbance – not polycythemia.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

Q39. Answer: B — Hypocalcemia (ionized calcium reduction) causing cardiac depression

Citrate (anticoagulant in stored blood) chelates ionized calcium, causing hypocalcemia which can lead to cardiac depression and coagulopathy.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

Q40. **Answer: B** **— Red cell membrane Na-K ATPase pump fails during storage, leaking potassium into plasma**

During storage, the RBC membrane Na-K ATPase pump becomes inactive, causing potassium to leak from cells into the plasma of stored blood.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

Q41. **Answer: B** **— Antibodies against donor leukocyte antigens or cytokines in stored blood**

Febrile non-hemolytic reactions are caused by recipient antibodies against donor WBC antigens or accumulated cytokines in stored blood components.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

Q42. **Answer: B** **— Pulmonary edema, hypertension, dyspnea due to volume overload**

TACO presents with signs of circulatory overload: pulmonary edema, raised JVP, hypertension, and dyspnea, especially in elderly and cardiac patients.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

Q43. **Answer: D** **— Tuberculosis (commonly)**

Transfusion-transmitted infections include hepatitis B, C, HIV, malaria, syphilis, and CMV. TB is not a common transfusion-transmitted infection.

Topic: Complications of Blood Transfusion | Ref: Ch 7, Page 77

Q44. **Answer: B** **— Transfusing the patient's own previously collected or salvaged blood**

Autologous transfusion involves collecting, storing, and re-infusing the patient's own blood, eliminating risks of alloimmunization and transfusion-transmitted infections.

Topic: Autologous Blood Transfusion | Ref: Ch 7, Page 79

Q45. **Answer: D** **— Emergency crossmatch from blood bank**

Autologous methods include preoperative donation, acute normovolemic hemodilution, and intraoperative cell salvage. Emergency crossmatch uses donor (allogeneic) blood.

Topic: Autologous Blood Transfusion | Ref: Ch 7, Page 79

Q46. **Answer: B** **— Crystalloids or colloids to maintain normovolemia**

In ANH, blood is withdrawn immediately before surgery and replaced with crystalloid/colloid to maintain intravascular volume. The blood is re-infused after surgical bleeding.

Topic: Autologous Blood Transfusion | Ref: Ch 7, Page 79

Q47. **Answer: B** **— Surgery for malignancy (relative) and infected surgical fields**

Cell salvage is relatively contraindicated in malignancy (risk of tumor cell dissemination) and contaminated/infected surgical fields.

Topic: Autologous Blood Transfusion | Ref: Ch 7, Page 79

Q48. **Answer: B** **— 7–8 g/dL**

In otherwise healthy patients, transfusion is generally triggered at hemoglobin 7–8 g/dL (restrictive strategy). Higher thresholds apply for cardiac disease.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

Q49. Answer: B — CPDA-1 (Citrate-Phosphate-Dextrose-Adenine)

CPDA-1 is the standard anticoagulant-preservative used in blood bags. Citrate prevents coagulation, phosphate and dextrose support RBC metabolism, and adenine maintains ATP levels.

Topic: Blood Transfusion | Ref: Ch 7, Page 75

Q50. Answer: B — 170–260 micrometers (standard blood filter)

Standard blood transfusion sets have a 170–260 micrometer filter to remove clots and debris. Microaggregate filters (20–40 µm) are used for special indications.

Topic: Blood Transfusion | Ref: Ch 7, Page 75